Spin-Orbit Torque and Nernst Effects in Bi_xSb_{1-x}/Ferromagnet Heterostructures

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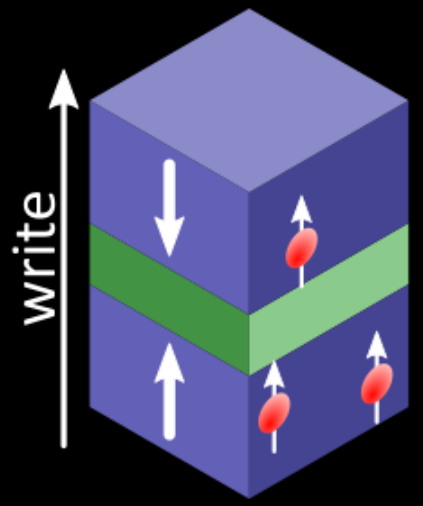
Spin-Torque Memory

In white boxed I will give comments to supplement the information on the slides.

Spin-Torque Memory

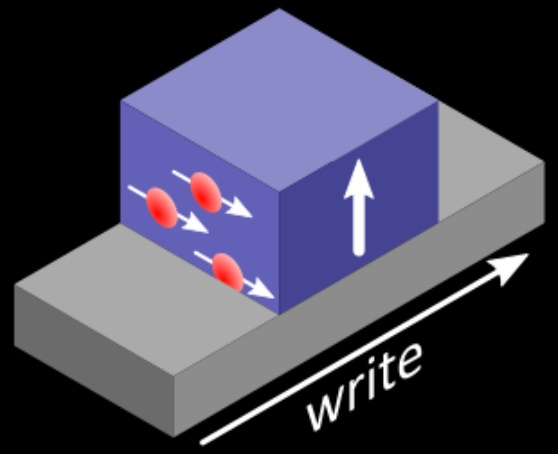
Spin-Transfer Torque - MRAM



- Fast
- Reliable
- High Efficiency

Spin-Orbit Torque - MRAM

- Faster
- Better Reliability
- Low Efficiency

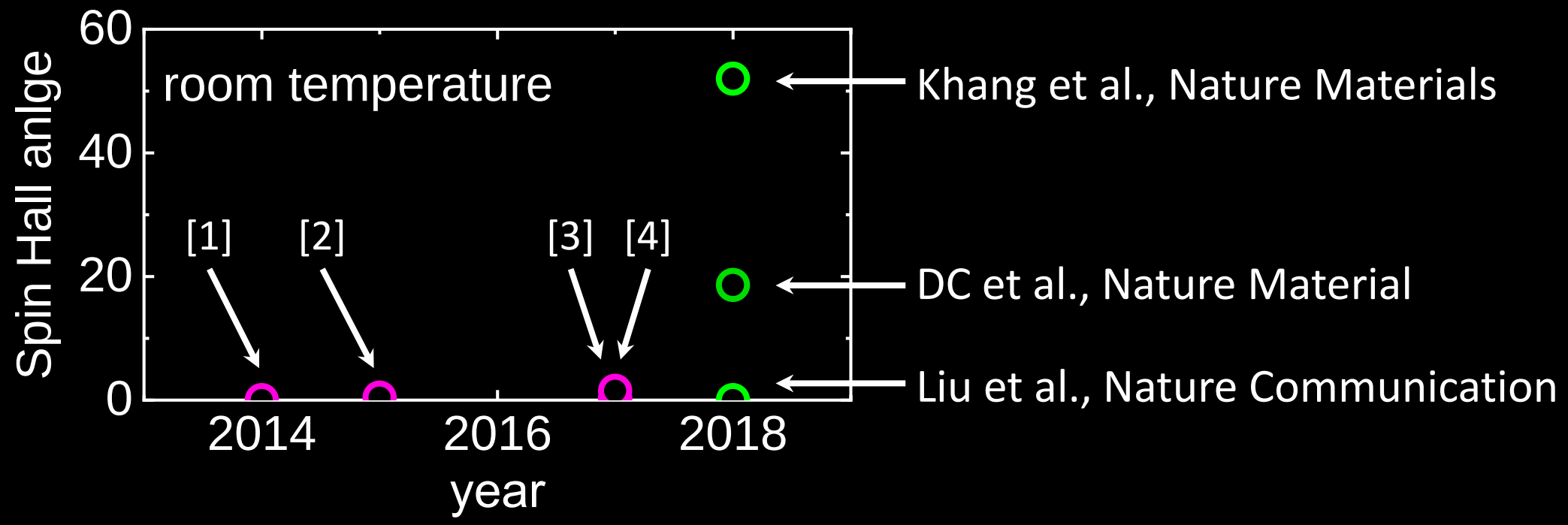


Is low SOT efficiency a materials challenge?

Spin Hall Angle in Topological Insulators

In 2018 two groups reported large spin Hall angles in topological insulator samples at room temperature. Specifically, Khang et al. observed a spin Hall angle of 52 in Bi-Sb using current switching experiments and DC et al. observed a spin Hall angle of 16 in sputtered Bi-Se using harmonic Hall measurements.

Spin Hall Angle in Topological Insulators



[1] Deorani et al., Physical Review B
[2] Jamali et al., Nano Letters
[3] Han et al., Physical Review Letters
[4] Wang et al., Nature Communications

Ordinary Nernst Effect in Topological Insulators

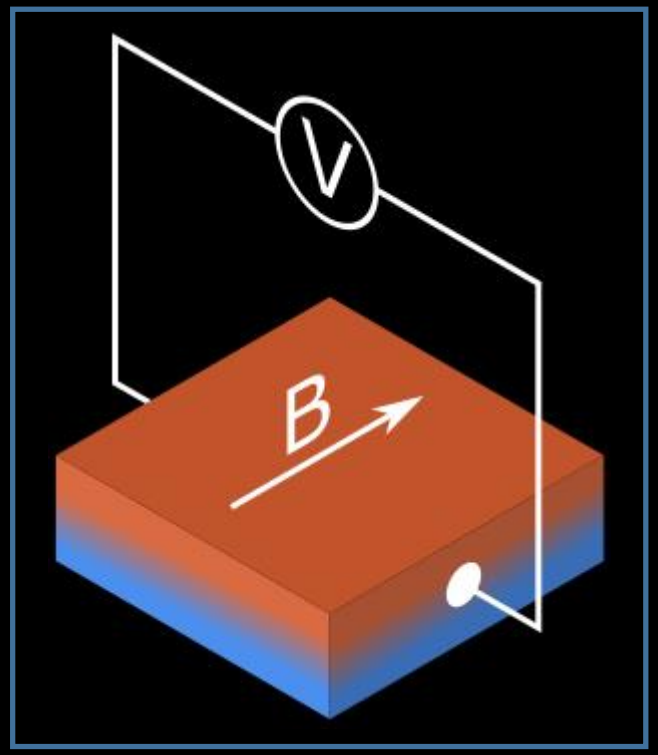
In this talk we will show harmonic Hall measurements in Bi-Sb/ferromagnet samples. However, we are not able to reproduce the results shown earlier. Instead we observe a large ordinary Nernst effect. Here, ordinary Nernst effect refers to the generation of a voltage perpendicular to an applied temperature gradient and magnetic field.

Since the ordinary Nernst effect has a similar signature in harmonic Hall measurements as spin-orbit torque, the spin Hall angle might have been overestimated in previous experiments.

Ordinary Nernst Effect in Topological Insulators

The ordinary Nernst effect is large in Topological Insulators

This might lead to an overestimation of the Spin Hall Angle



Overview

In the following we will show a set of three control experiments to investigate the origin of the harmonic Hall signal in Bi-Sb/Co samples:

1. The harmonic Hall voltage does not vanish, if we replace the magnet with aluminum. Therefore the harmonic Hall signal originates from a (thermal) effect in the Bi-Sb and not from spin-torque at the Bi-Sb/Co interface
2. The harmonic Hall voltage scales linear with the external magnetic field. This is expected from the ordinary Nernst effect but contradicts the theory of spin-orbit torque.
3. The electric field generated in the longitudinal and transverse direction are equal. This indicated that the electric fields in both measurements has the same physical origin. However, SOT can only be observed in the transverse voltage.

Harmonic Hall voltage persists in control samples without magnet

Harmonic Hall voltage is not suppressed at large magnetic field

A longitudinal electric field of same magnitude is generated

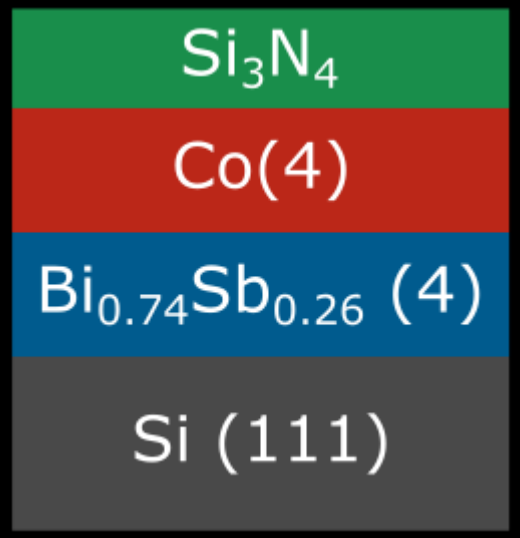
Sample Overview

In the following we will compare 3 different samples:

1. Bi-Sb/Co was expected to show large spin-orbit torque. In addition thermal effects can be observed in the second harmonic Hall voltage.
2. No spin-orbit torque can be observed in Bi-Sb/Al due to the absence of a magnet. However, thermal effect will still persist.
3. Pt/Py is a reference sample with detectable spin-orbit torque . Thermal effects are small, because temperature gradient in metallic layers are small.

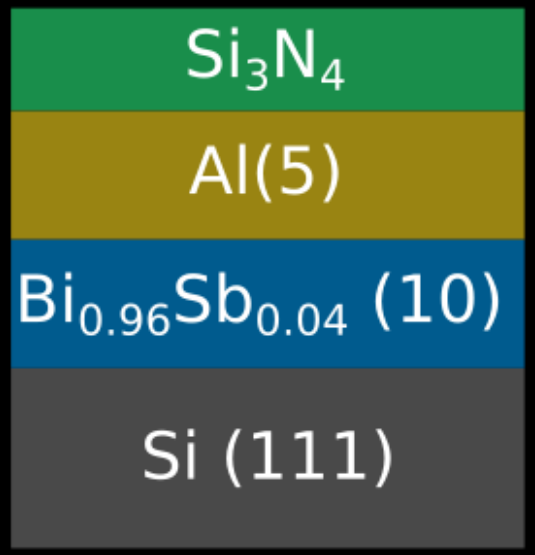
Sample Overview

Bi-Sb/Co



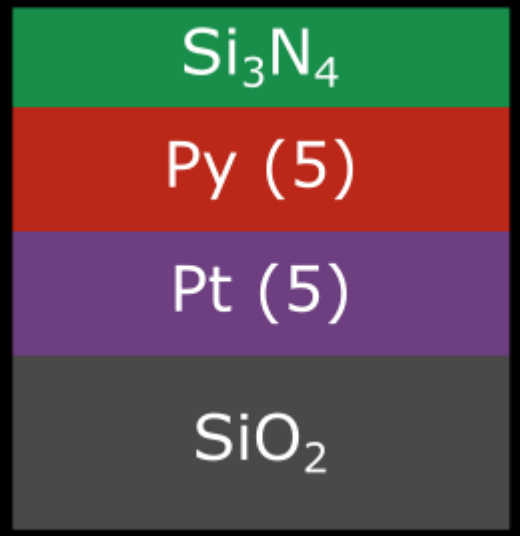
- Spin-Orbit Torque
- Thermal Effects

Bi-Sb/Al



- No Spin-Orbit Torque
- Thermal Effects

Pt/Py

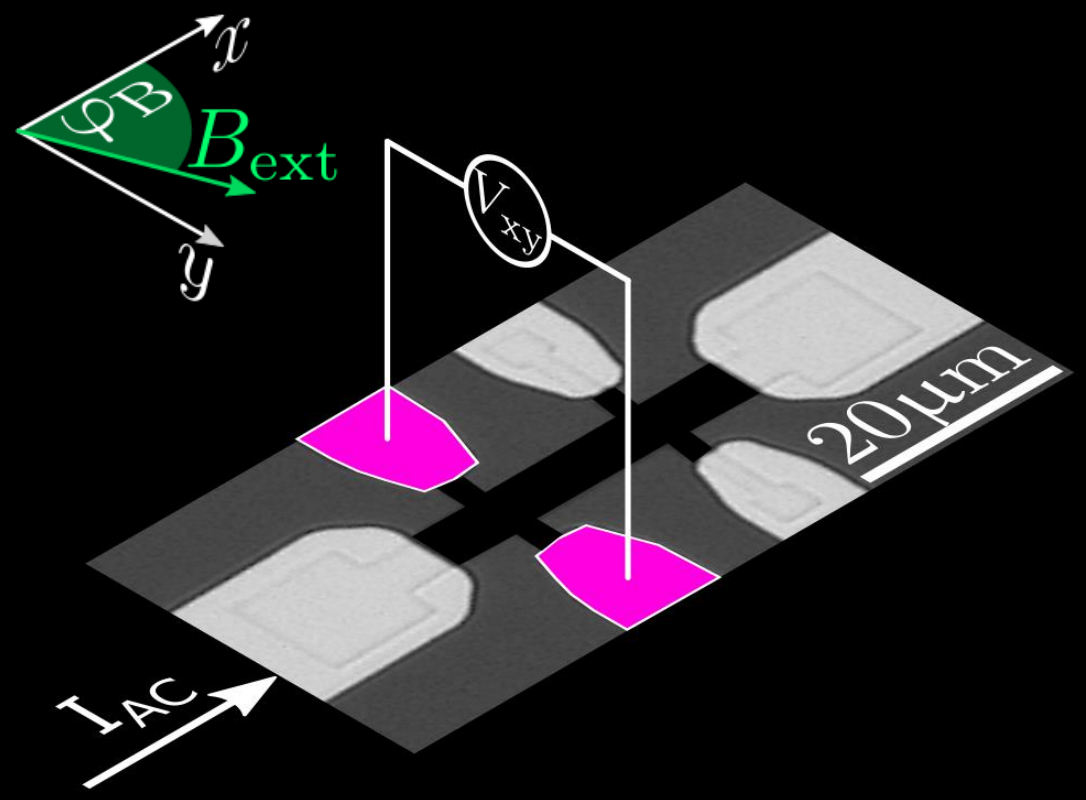


- Spin-Orbit Torque
- Small Thermal Effects

Harmonic Hall Measurements of SOT

For harmonic Hall measurements double Hall bars are patterned. Both the first and second harmonic Hall voltage is measured. We expect to observe the planar Hall effect in the first harmonic signal if a magnet is present.
The second harmonic Hall voltage will show damping-like spin-orbit torque “cos(φ)”, field-like spin-orbit torque “cos(2φ)cos(φ)” and thermal effects (ordinary/ anomalous Nernst effect “cos(φ)”)

Harmonic Hall Measurements of SOT



Hall Voltage V_{xy}

First Harmonic Hall Voltage

- Planar Hall Effect $\sin(2\varphi_B)$

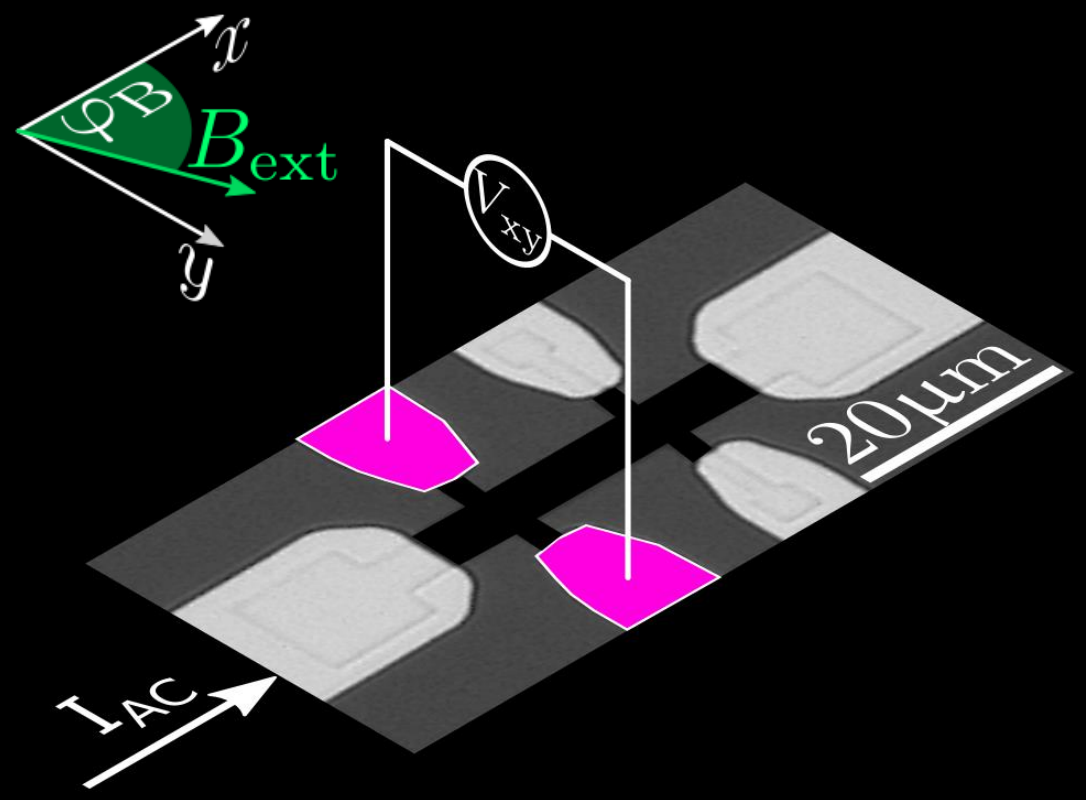
Second Harmonic Hall Voltage

- Damping-like Spin-Orbit Torque $\cos(\varphi_B)$
- Field-Like Spin-Orbit Torque
- Thermal Effects $\cos(\varphi_B)$

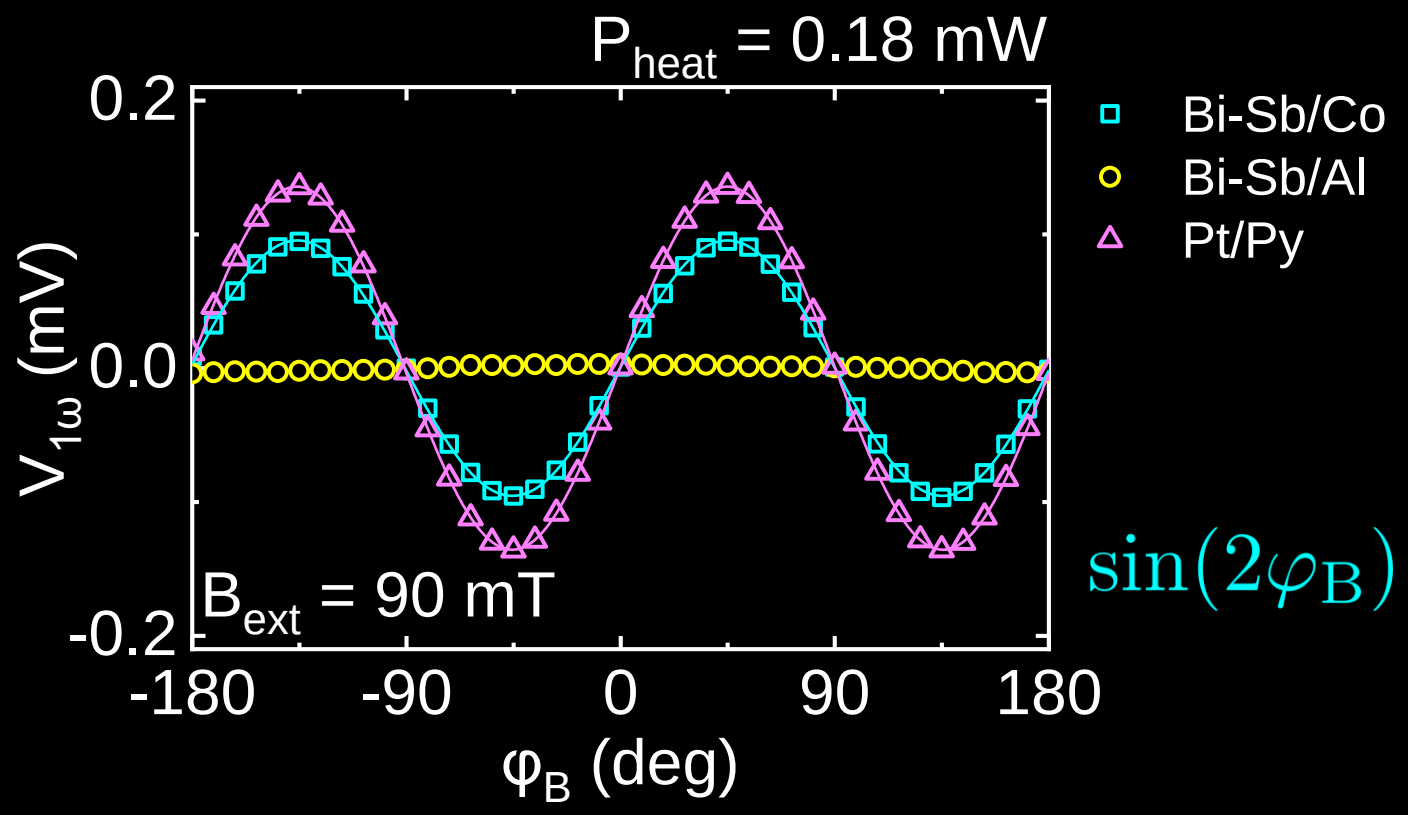
Harmonic Hall Measurements of SOT

The first harmonic Hall voltage shows a sin(2φ) angular dependence, expected from the planar Hall effect. In the sample Bi-Sb/Al, the signal is absent, since no magnet is present.

Harmonic Hall Measurements of SOT



Hall Voltage V_{xy}



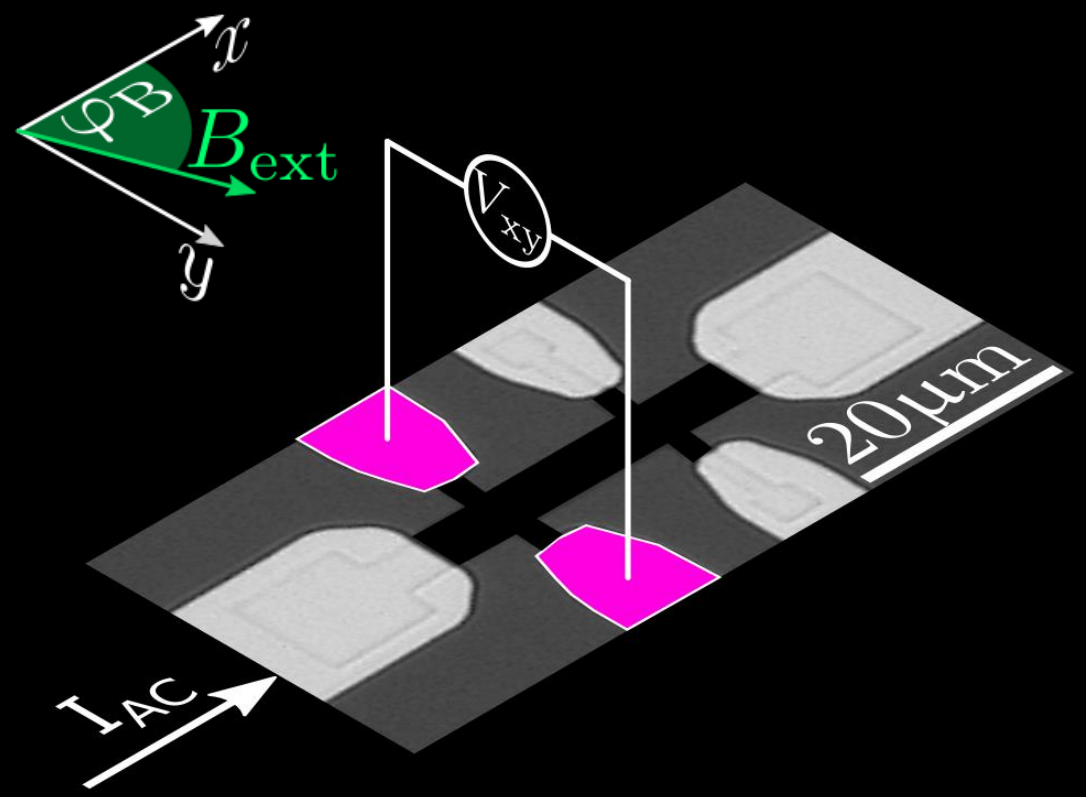
Harmonic Hall Measurements of SOT

In the second harmonic Hall voltage, a large signal is observed in the sample Bi-Sb/Co. The cos(φ) angular dependence could be interpreted as damping-like SOT. However, the Nernst effect would show the same angular dependence.

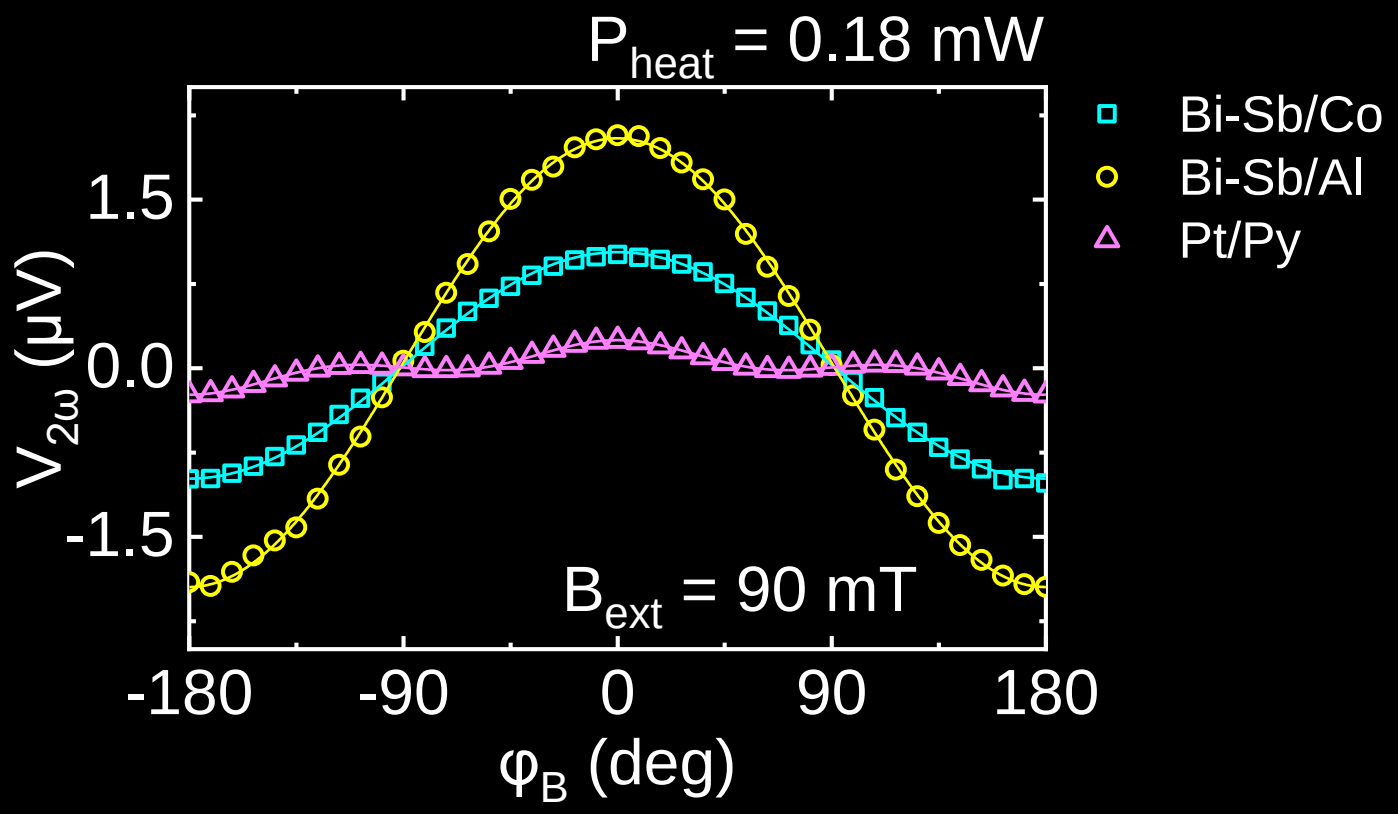
In the sample Bi-Sb/Al, the large signal persists despite the absence of a magnet. This indicated that the origin of the large signal is a thermal effect.

Finally, a more complex angular dependence is observed in the sample Pt/Py due to the presence of a large field-like spin-orbit torque. The amplitude of the damping-like spin-orbit torque is small in comparison to the signal is the Bi-Sb sample.

Harmonic Hall Measurements of SOT



Hall Voltage V_{xy}



Harmonic Hall voltage persists in control samples without magnet

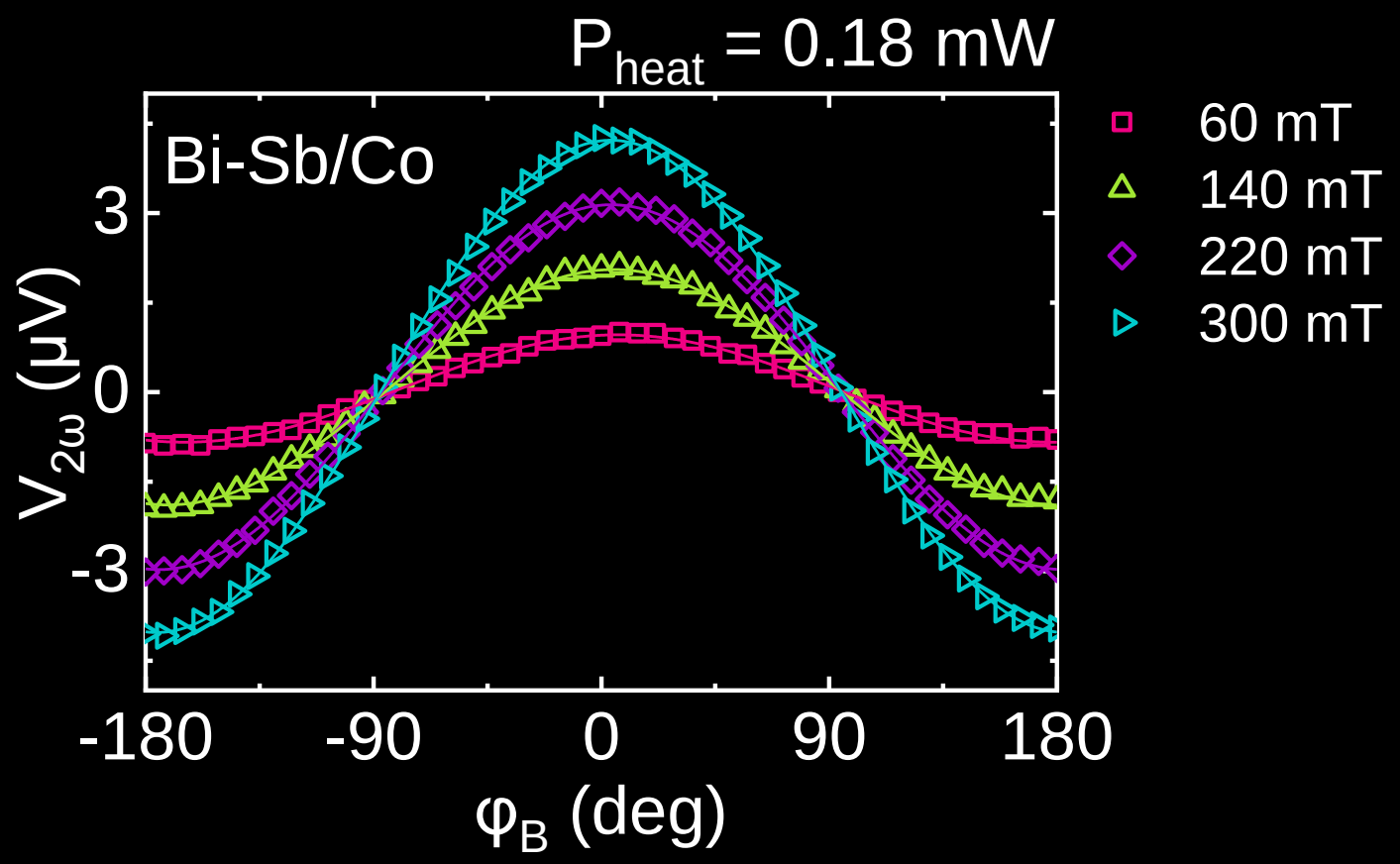
Harmonic Hall voltage is not suppressed at large magnetic field

A longitudinal electric field of same magnitude is generated

Magnetic Field Scaling

Damping-like spin-orbit torque is expected to scale as $(B_{ext}+\mu_0 M_{eff})^{-1}$, while the ordinary Nernst effect is linear in magnetic field. From our measurements we conclude that the signal in the sample Bi-Sb/Co is dominated by the ordinary Nernst effect and not spin torque.

Magnetic Field Scaling



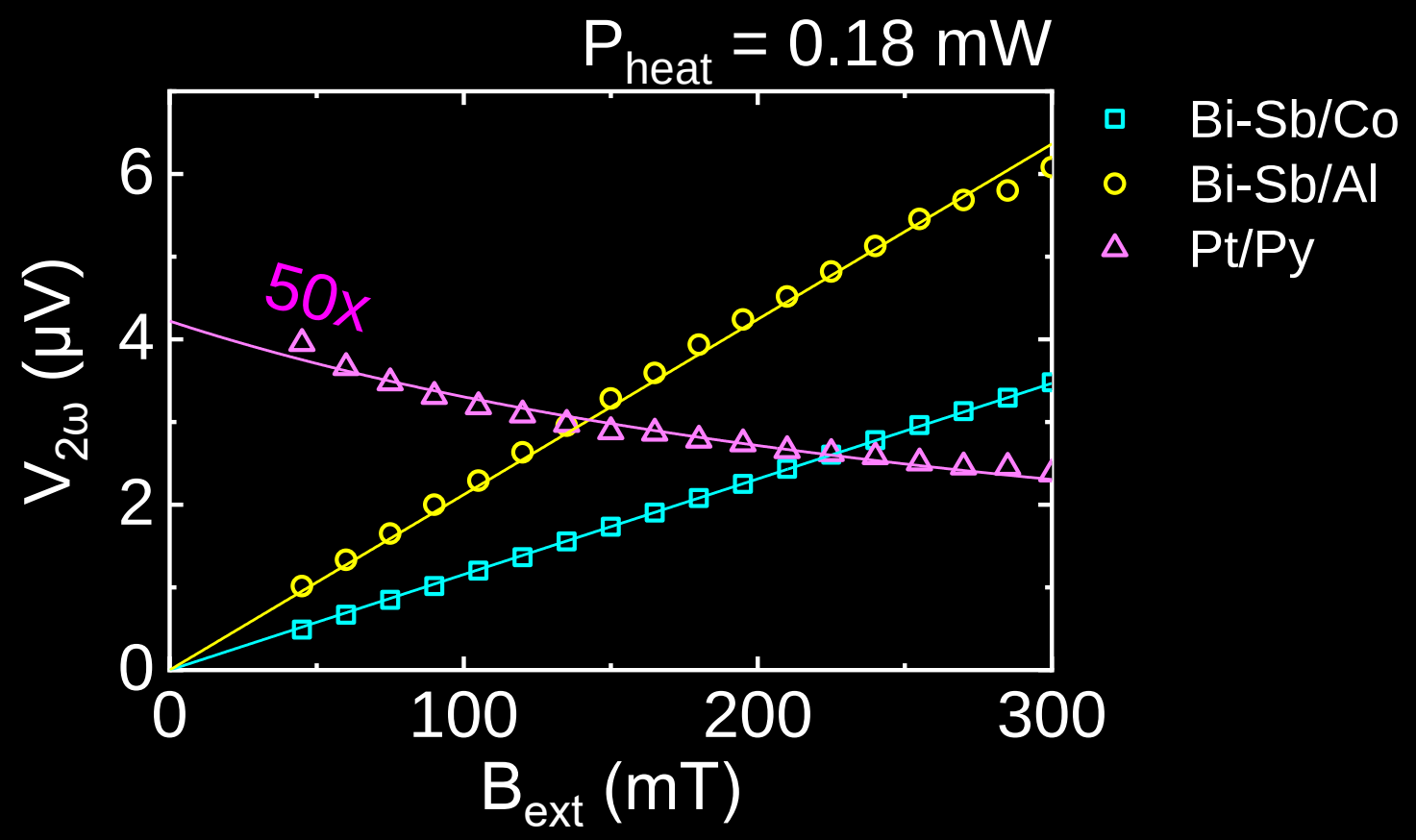
Ordinary Nernst Effect

$$V_{2\omega} \propto B_{\text{ext}}$$

Damping-Like Spin-Orbit Torque

$$V_{2\omega} \propto \frac{1}{B_{\text{ext}} + \mu_0 M}$$

Magnetic Field Scaling



Linear scaling confirms presence of ordinary Nernst effect

Harmonic Hall voltage persists in control samples without magnet

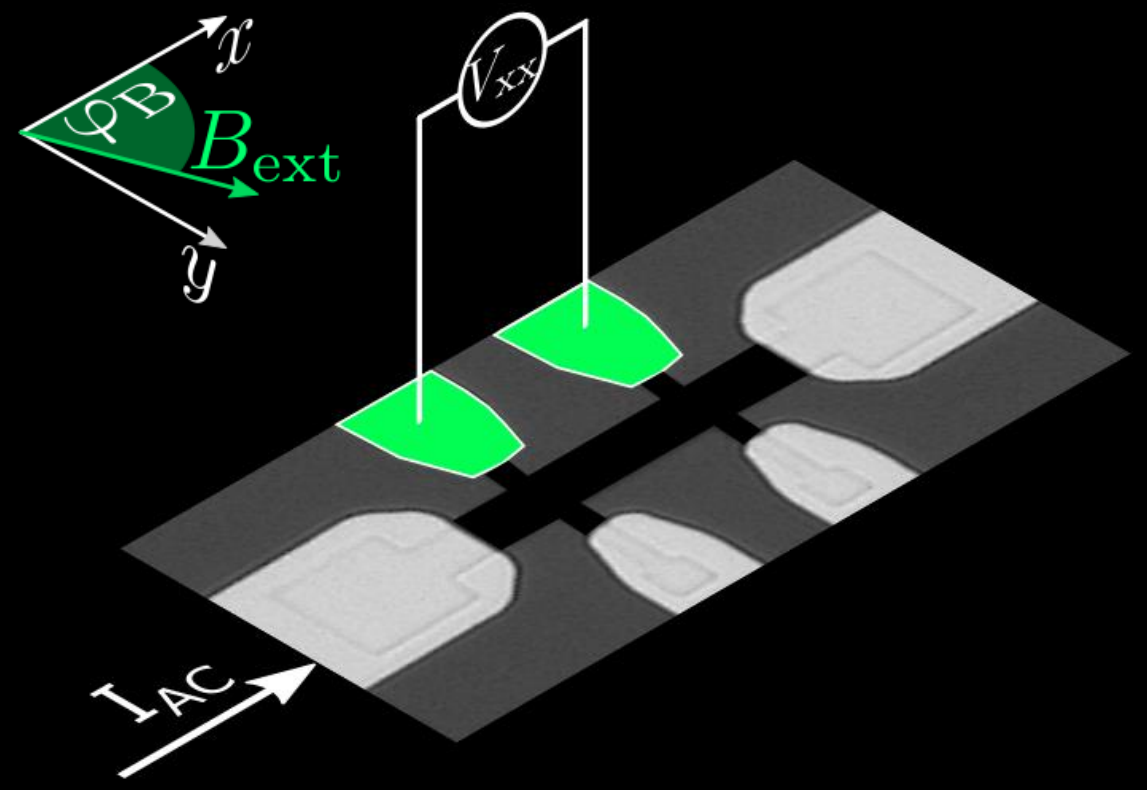
Harmonic Hall voltage is not suppressed at large magnetic field

A longitudinal electric field of same magnitude is generated

Longitudinal Voltage

In addition to the transverse voltage we study the longitudinal voltage. If we do experiments where we apply an in-plane magnetic field at an angle φ_B, we expect to observe anisotropic magnetoresistance in the first Harmonic voltage, while the second harmonic Hall voltage will show signatures of field-like spin-orbit torque ("cos(2φ)sin(φ)"), unidirectional spin-Hall magneto-resistance ("sin(φ)") and Nernst effect ("sin(φ)").

Longitudinal Voltage



First Harmonic Hall Voltage

- Anisotropic Magnetoresistance $\cos(2\varphi_B)$

Second Harmonic Hall Voltage

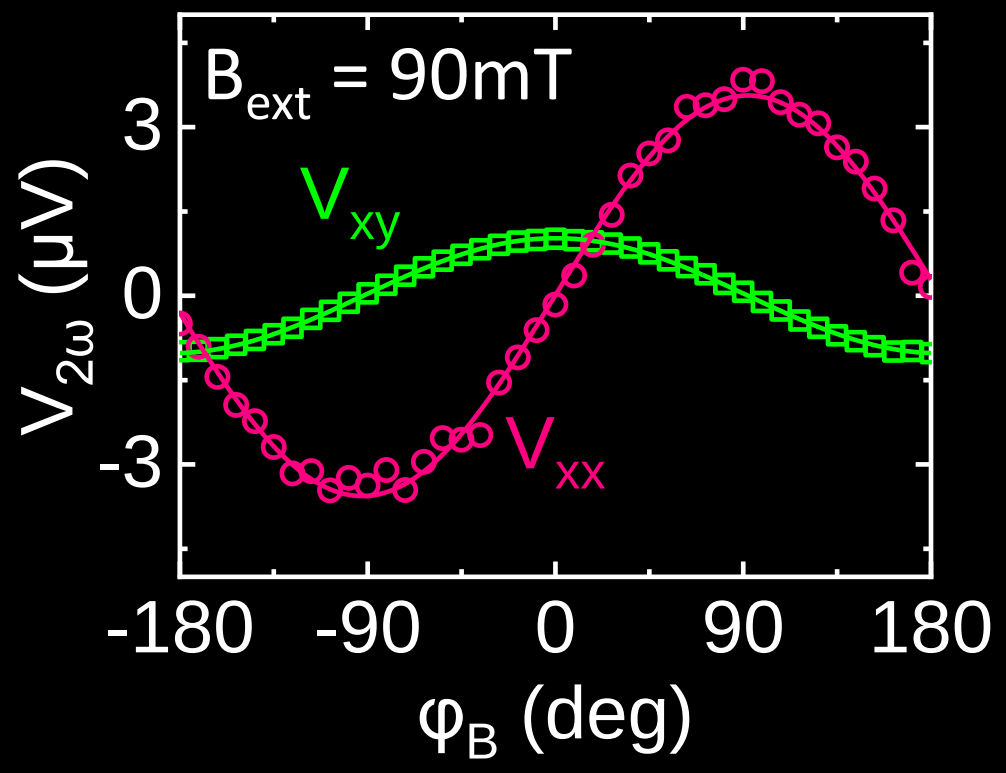
- Unidirectional Spin Hall MR $\sin(\varphi_B)$
- Thermal Effects $\sin(\varphi_B)$
- No Damping-Like SOT

Nernst Effect in Bi-Sb/Co

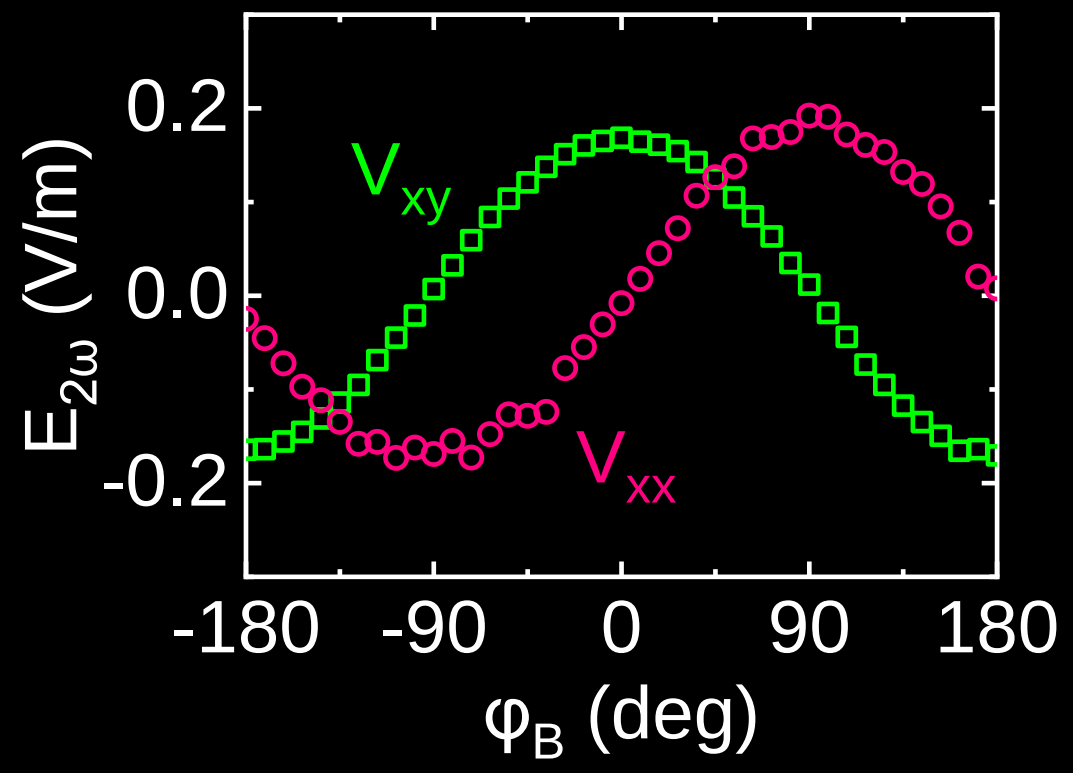
We observe the same electric field in the longitudinal and transverse direction. This indicates that both signals have the same physical origin. This is expected if a large ordinary Nernst effect is present. On the other hand, spin-orbit torque and unidirectional spin-Hall magnetoresistance are distinctly different effects. Thus it is not expected that the electric field stemming from USMR and damping-like SOT would be equal in magnitude.

Nernst Effect in Bi-Sb/Co

Voltage

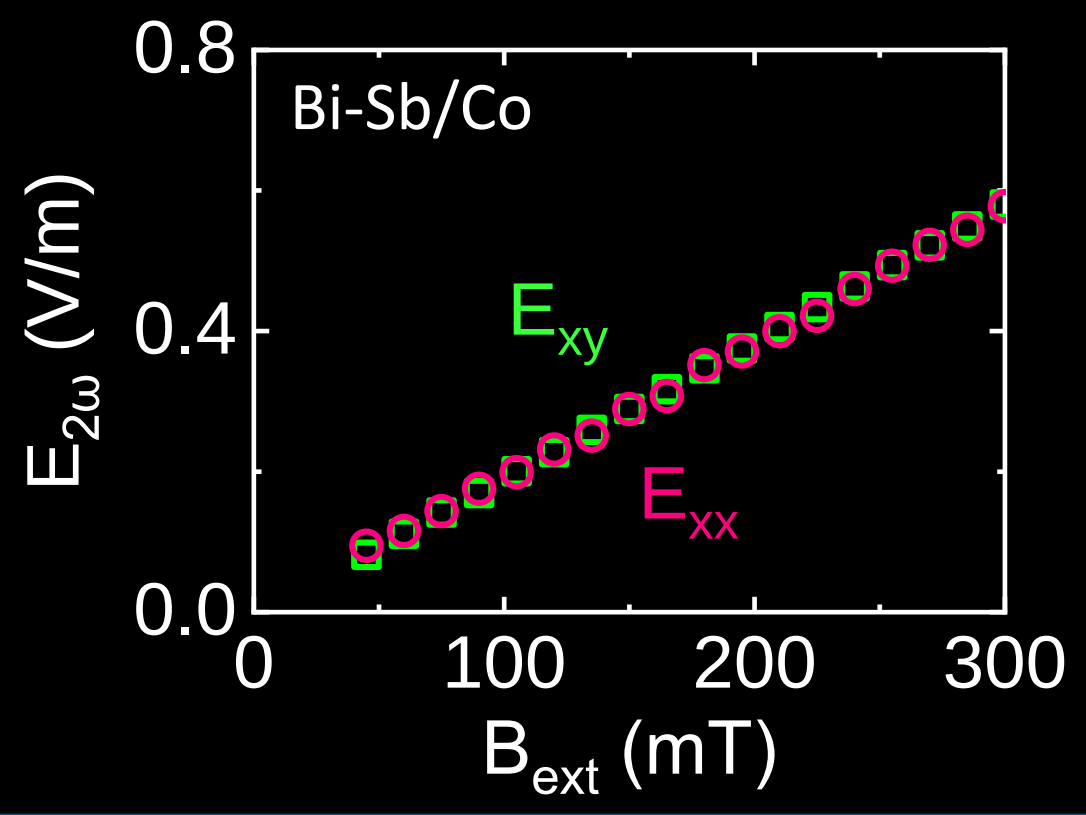


Electric Field

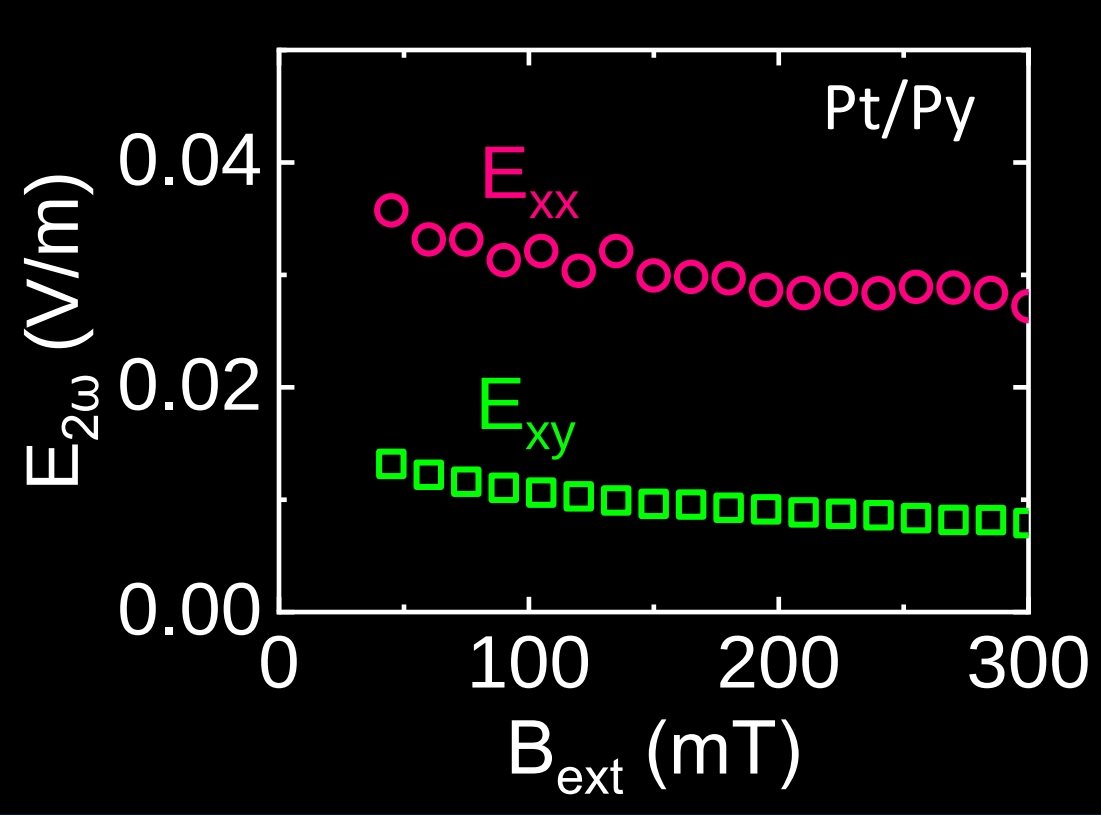


Spin Orbit Torque - Electric field

Ordinary Nernst Effect

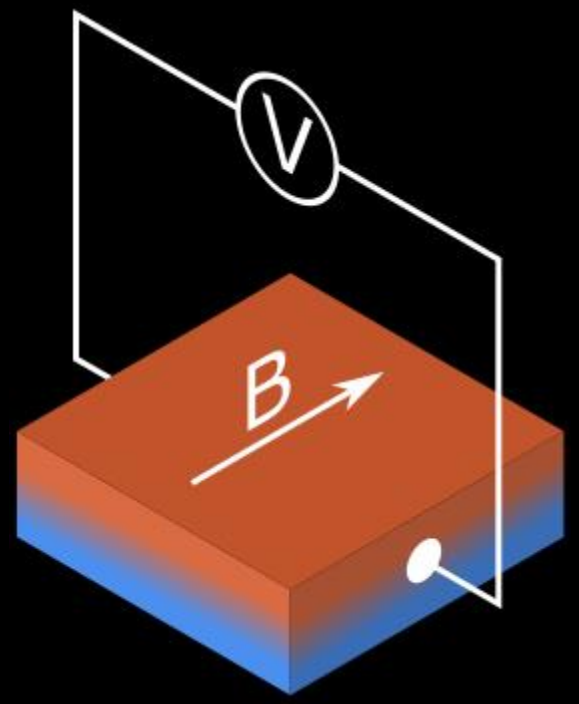


Spin-Orbit Torque & USMR



Conclusion

1. Nernst Effect can be large in semiconducting samples
2. Topological Insulators might not be an ideal material for MRAM applications



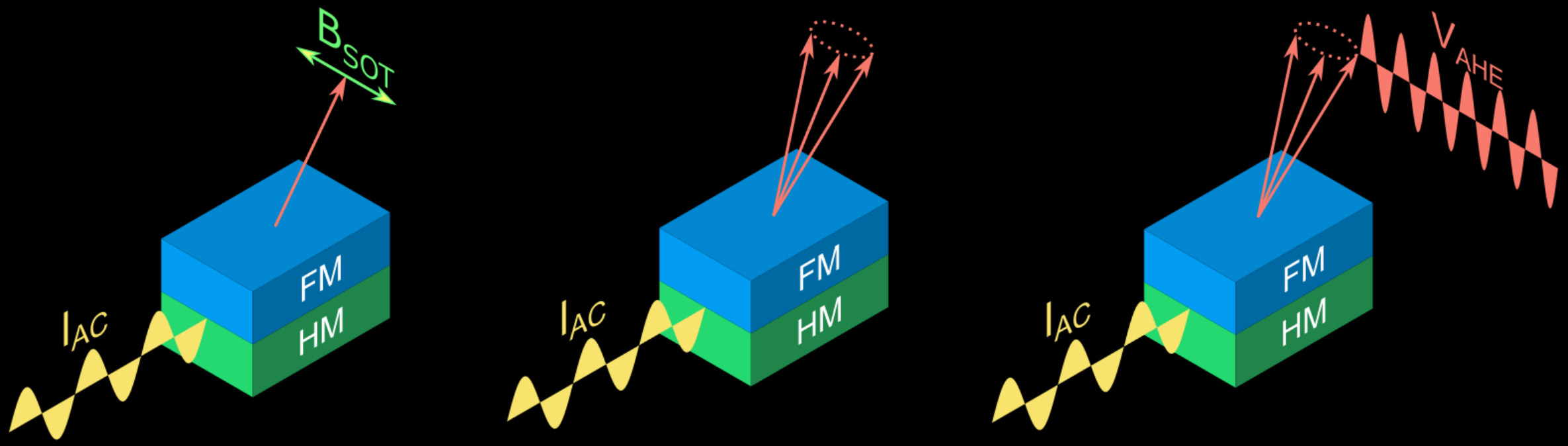
Access Presentation: www.roschewsky.com

Appendix

Nernst Effect in Bi-Sb/Co

An ac charge current induces an ac spin accumulation at the spin-Hall metal / magnet interface at the same frequency. Due to the presence of spin-orbit torque, an alternating effective magnetic field is action on the magnetization. This causes quasi-static oscillations of the magnetization. Due to the presence of the anomalous Hall effect, the resistance of the sample chances at the same frequency as the ac current. Therefore the voltage changes at twice the frequency.

Second Harmonic Hall Voltage

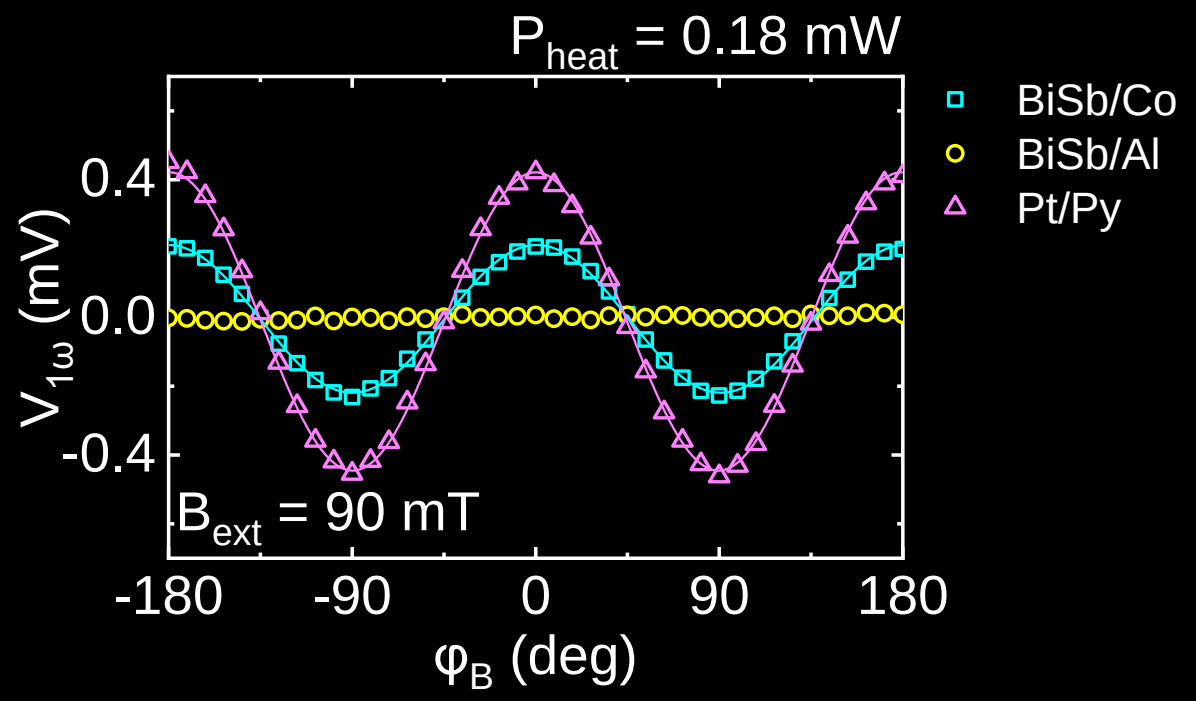


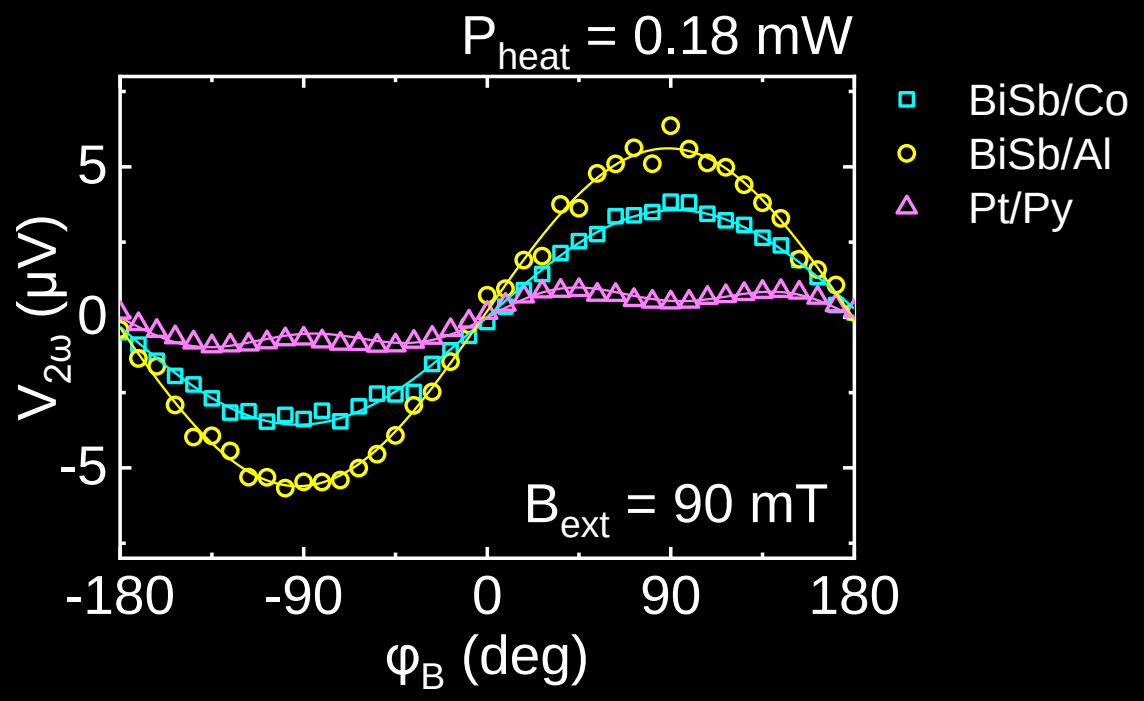
I_{AC} induces B_{SOT} → M oscillates with B_{SOT} → V_{xy} changes periodically

Anisotropic Magnetoresistance and USMR

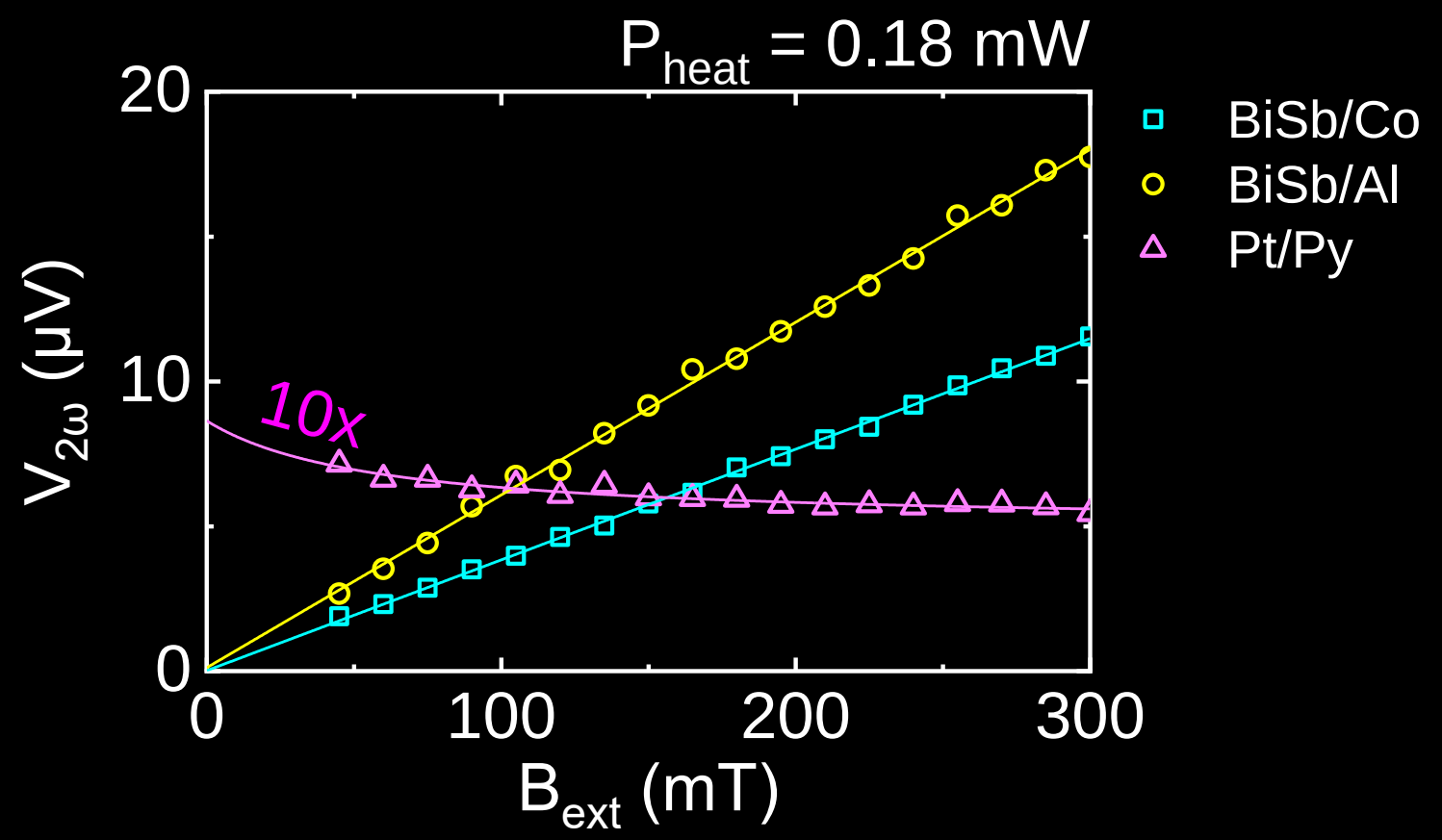
Supplementary data showing the AMR and USMR/Nernst effect in all samples studied.

Anisotropic Magnetoresistance and USMR





Anisotropic Magnetoresistance and USMR

Scaling of the longitudinal second harmonic Hall voltage with magnetic field. The linear trend is indicative of ordinary Nernst effect. In the Pt/Py sample, an inverse trend can be observed. This is expected from the unidirectional spin-Hall magneto-resistance.

Anisotropic Magnetoresistance and USMR